User Manual

SentiAI — Amazon Review Sentiment Analyzer

0 Welcome to SentiAI

SentiAI reads Amazon product review text and instantly tells you whether the review expresses a **positive** or **negative** sentiment. It uses a deep learning model trained on thousands of real Amazon reviews to understand the tone and meaning of text — not just individual words.

What you can do with SentiAI:

- Analyze a single review to see if it is positive or negative
- Analyze up to 50 reviews at once (batch mode)
- View your recent prediction history
- Check the status of the AI pipeline

1 Getting Started

1.1 Step 1 — Open the Application

Open your web browser and navigate to:

```
http://localhost:3002
```

You will see the SentiAI home screen with a dark background and a text input area.

1.2 Step 2 — Understanding the Screen Layout

The application has two main sections accessible from the top navigation bar:

Table 1: Navigation Bar Buttons

Button	What it does
Analyze	The main page for analyzing reviews
ML Pipeline	Shows the status of the AI system

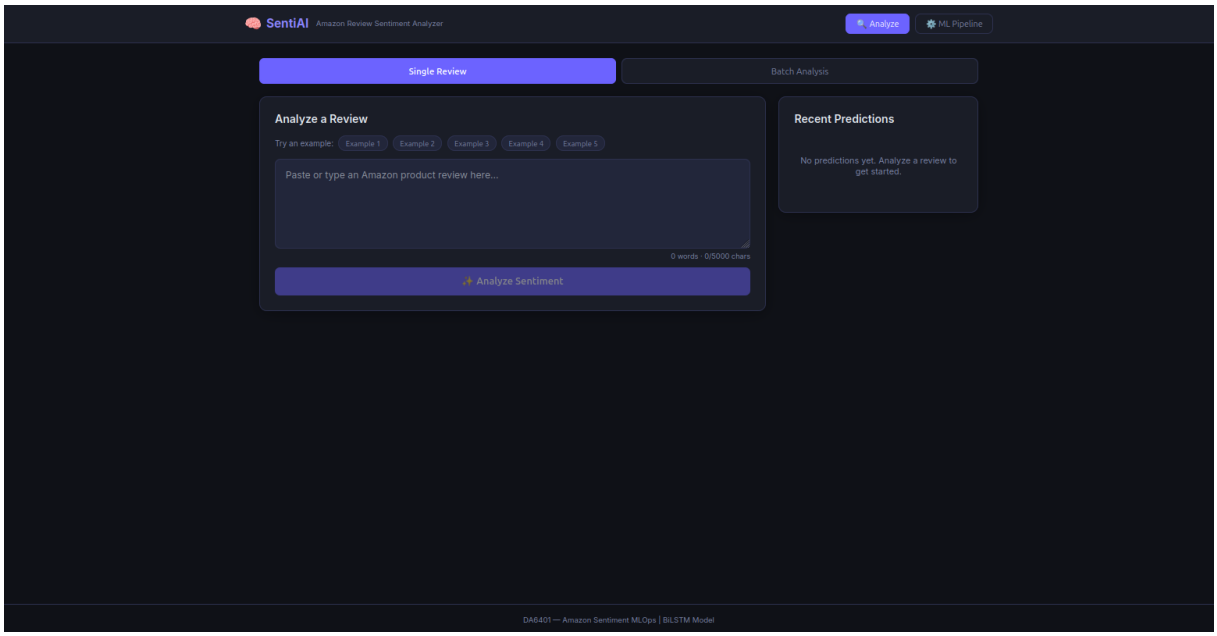

Figure 1: Analyze

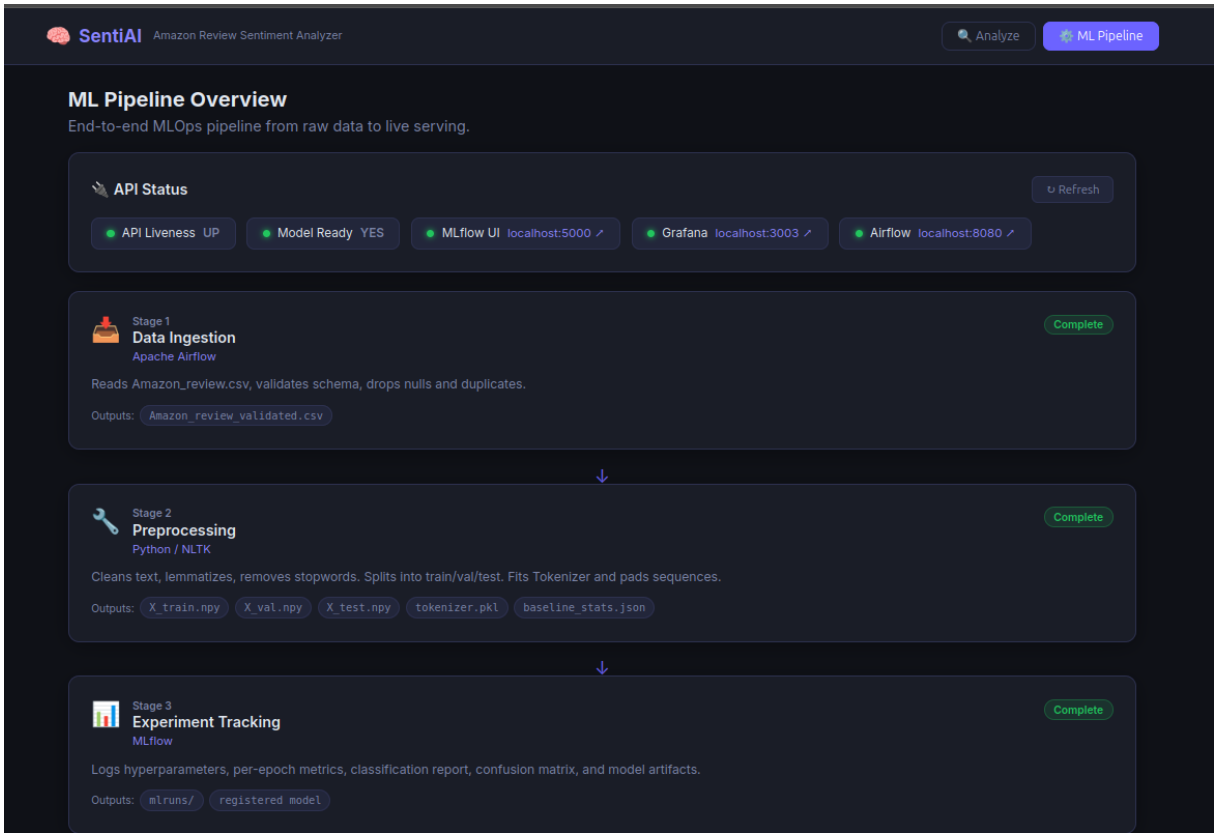

Figure 2: ML Pipeline

2 Page 1: Analyze Reviews

This is the main page where you analyze review text.

2.1 Part A — Single Review Analysis

How to analyze one review:

Step 1. Click inside the large text box that says *"Paste or type an Amazon product review here..."*

Step 2. Type or paste the review text you want to analyze.

Step 3. Click the **"Analyze Sentiment"** button.

Step 4. Wait 1–2 seconds — a result card will appear below the button.

Example:

> *"This product is absolutely fantastic. Arrived quickly, works perfectly, and the quality is outstanding. Highly recommend to anyone!"*

Click Analyze → Result shows **POSITIVE — 100% confidence**.

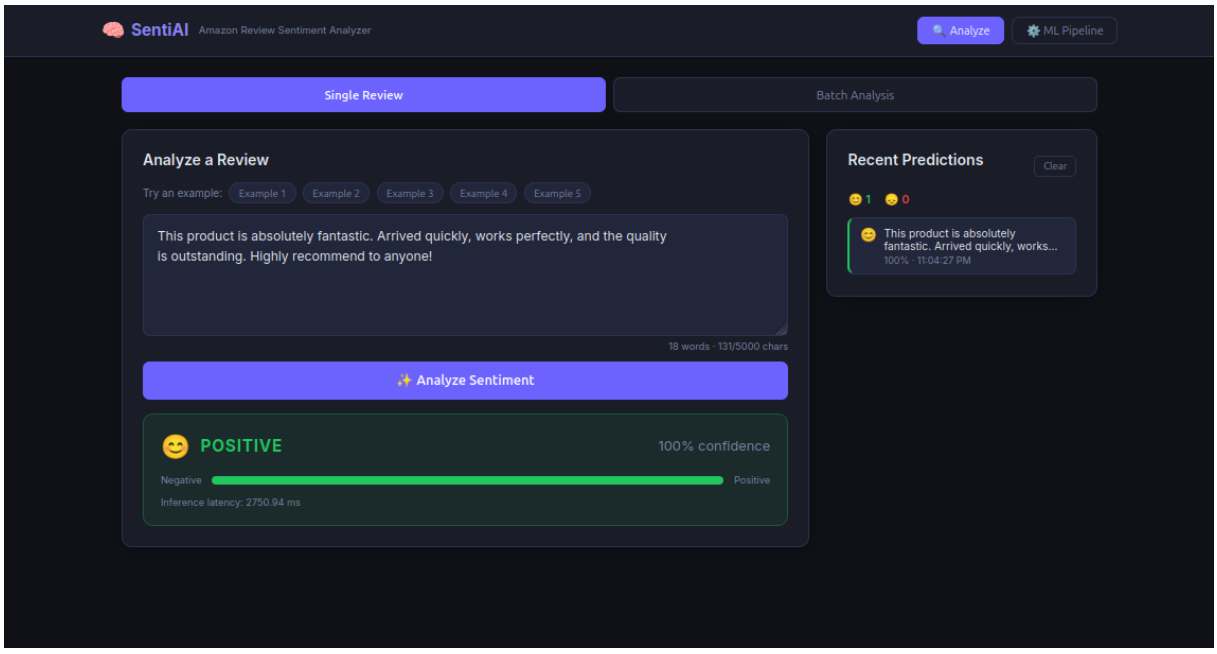

Figure 3: Analyze

2.2 Understanding the Result Card

When a prediction appears, you will see the following elements:

Table 2: Result Card Elements

Element	What it means
Sentiment icon	The predicted sentiment — a positive or negative indicator
Percentage	How confident the model is in its prediction (higher = more certain)
Colored bar	Visual split between negative probability (left) and positive probability (right)
Inference latency	How fast the analysis was completed, in milliseconds

Note

A confidence of around 50% means the review is ambiguous — the model is uncertain. Reviews that are clearly positive or negative will have higher confidence scores.

2.3 Using the Example Buttons

Not sure what to type? Click any of the **Example 1, 2, 3...** buttons at the top of the text box. This automatically fills the input with a sample review so you can see how the analyzer works without typing anything.

2.4 Part B — Batch Analysis (Multiple Reviews at Once)

When to use batch mode:

- When you have many reviews to check quickly
- When you want to see an overall positive/negative breakdown

How to use batch mode:

Step 1. Click the **"Batch Analysis"** tab at the top of the page.

Step 2. Paste your reviews into the text box — **one review per line**.

Step 3. You can enter up to **50 reviews** at a time.

Step 4. Click the **"Analyze N Reviews"** button.

Step 5. Results appear below with a summary and individual predictions.

Example input (one review per line):

```
Great product, very happy with this purchase!
Absolutely terrible, stopped working after a week.
Average quality, nothing special but does the job.
```

After analysis, you will see a summary like:

1 Positive 2 Negative 33% positive rate

This is followed by individual result cards for each review.

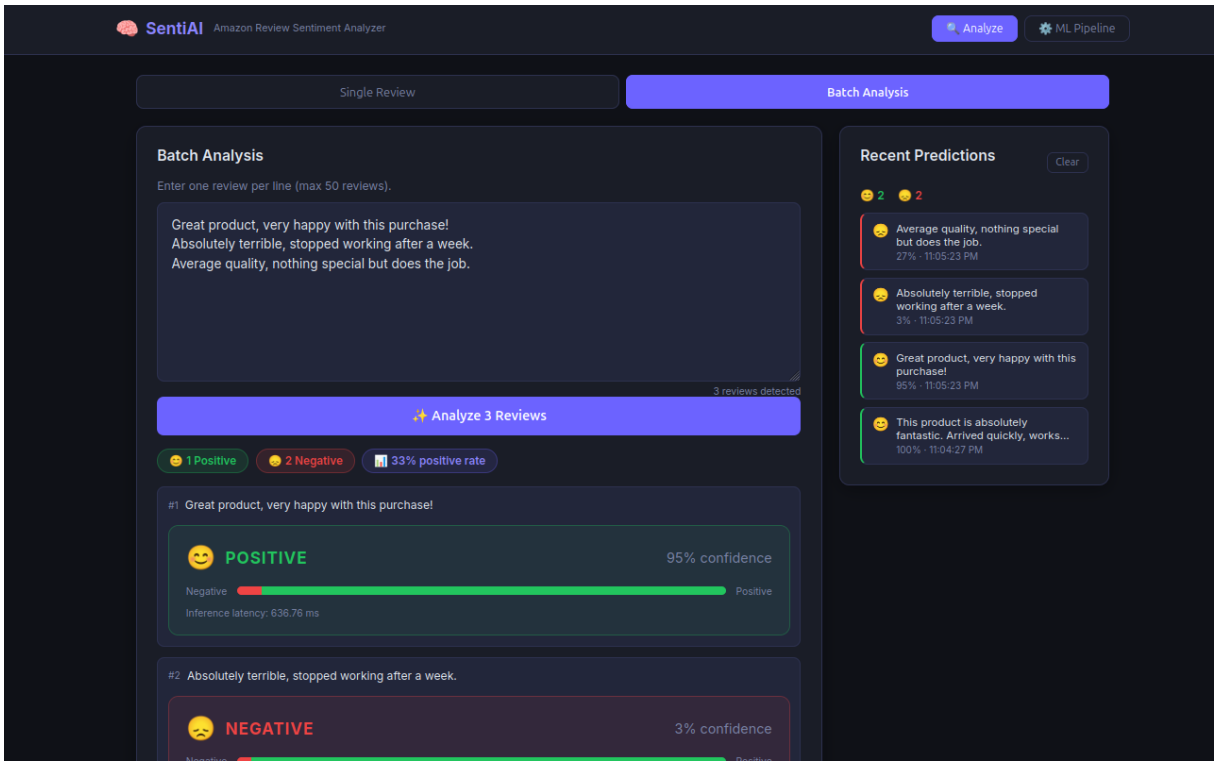

Figure 4: Analyze

2.5 Part C — Recent Predictions Panel

On the right side of the Analyze page you will see a **Recent Predictions** panel. This keeps a history of your last 20 predictions during your current session.

What each item shows:

- The predicted sentiment (positive or negative)
- The first 60 characters of the review
- The confidence percentage
- The time the prediction was made

To clear the history: Click the **"Clear"** button at the top of the panel.

Note

The history disappears when you close or refresh the browser tab. It is not saved permanently.

3 Page 2: ML Pipeline

Click the **"ML Pipeline"** button in the top navigation bar to view this page.

3.1 API Status Panel

This panel shows whether the backend services are currently running and healthy.

Table 3: Status Indicators

Indicator	What it means
UP / YES	The service is running normally
DOWN / NO	The service is not responding
Refresh	Click to check the current status

Links to tool interfaces:

- `localhost:5000` — Opens the MLflow experiment tracking interface
- `localhost:3001` — Opens the Grafana monitoring dashboard
- `localhost:8080` — Opens the Airflow pipeline scheduler

4 What the Confidence Score Means

Table 4: Confidence Score Interpretation

Confidence	What it indicates
90% – 100%	Very strong signal — clearly positive or negative
70% – 90%	Strong signal — model is fairly certain
55% – 70%	Moderate signal — review has mixed signals
50% – 55%	Uncertain — the review is ambiguous or neutral

Note

Example of an ambiguous review:
"The product arrived on time and looks nice, but the quality is not quite what I expected for the price."

This has both positive and negative signals, so the confidence may be around 55–65%.

5 Troubleshooting

5.1 The page does not load at all

Cause: The application is not running.
Fix: Start the application with `docker compose up` and wait 60 seconds, then try again.

5.2 "Failed to fetch" error appears

Cause: The backend API server is not running or not reachable.
Fix:

1. Check that Docker is running on your computer.
2. Open a terminal and run: `docker compose ps`
3. All containers should show as `Up` and `healthy`.
4. If not, run: `docker compose up -d`

5.3 The result says "Model not ready"

Cause: The AI model is still loading into memory after startup.
Fix: Wait 20–30 seconds and click Analyze again. The model takes a moment to load when the application first starts.

5.4 Very low confidence score (close to 50%)

Cause: The review contains mixed positive and negative signals, or is too short for the model to analyze confidently.
Fix: Try a longer, more specific review. Single-word reviews like *"good"* or *"bad"* will always produce low confidence.

5.5 The app looks broken or unstyled

Cause: The browser may have cached an old version.
Fix: Try a hard refresh:

- Windows/Linux: `Ctrl + Shift + R`
- Mac: `Cmd + Shift + R`

5.6 Batch mode says "0 reviews detected"

Cause: The reviews may not be separated by line breaks.
Fix: Make sure each review is on its own line. Press `Enter` after each review before clicking Analyze.

6 Glossary

Table 5: Glossary of Terms

Term	Plain English explanation
Sentiment	The overall feeling or opinion expressed in a piece of text — positive means the writer is happy, negative means they are unhappy
Confidence	How certain the AI model is about its prediction — higher is more certain
Threshold	The cut-off point the model uses to decide positive vs. negative — shown in the result card
Batch analysis	Analyzing multiple reviews at the same time instead of one by one
Inference	The process of the AI making a prediction
Latency	How long the prediction took to complete, measured in milliseconds
ML Pipeline	The sequence of steps the AI system uses to process data and make predictions
API	The communication layer between the frontend (what you see) and the AI model (the backend)

7 Quick Reference Card

SentiAI Quick Reference

Open app: `http://localhost:3002`

Single review:

1. Paste review in text box
2. Click **Analyze Sentiment**
3. See result below

Batch (up to 50):

1. Click **Batch Analysis** tab
2. One review per line
3. Click **Analyze N Reviews**

Result means:

- **POSITIVE** — review expresses a positive sentiment
- **NEGATIVE** — review expresses a negative sentiment
- **%** — how confident the model is

If something breaks:

```
docker compose up -d
```

Wait 30 seconds, then refresh the browser.